

FLOATING POINT ARITHMETIC AND STABILITY

1.1 SOME DEFINITIONS

LET a BE SOME EXACT (REFERENCE) VALUE,
AND LET $\hat{a}$ BE SOME APPROXIMATION OF a (FLOATING
PT. APPROX)

ABSOLUTE ERROR (SOMETIMES SIMPLY ERROR)

$$e_{\text{abs}} = a - \hat{a}$$

(OR $\hat{a} - a$, DOESN'T MATTER)

RELATIVE ERROR (FUNCTIONAL ERROR)

$$e_{\text{REL}} = \frac{a - \hat{a}}{|a|} \quad \text{FOR } a \neq 0$$

1.2 IEEE 754 64-BIT FLOATING PT ARITHMETIC

- 53 BITS MANTISSA, 11 BITS EXPONENT, BASE 2 REPRESENTATION FOR BOTH; CAN REPRESENT NUMBERS IN THE RANGE

$$\sim 2 \times 10^{-308} \dots 1 \times 10^{308}$$

WITH TOUGHLY 16 DECIMAL DIGITS PRECISION

- $1/f_{\min} < f_{\max}$ SO THAT $1/f_{\min}$ DOESN'T OVERFLOW
- SPECIAL (EXCEPTIONAL) VALUES

infinity EXAMPLE: $1.0 / 0.0$

nan (NOT A NUMBER) EXAMPLE: $0.0 / 0.0$

- nan is "CONTAGIOUS" nan $[* / + -]$ ANYTHING = nan e.g.

1.3 MACHINE PRECISION / MACHINE EPSILON, ϵ

- SMALLEST $\epsilon > 0$ SUCH THAT

$$|t - c| \neq 1$$

IN THE FLOATING PT MODEL

$$\text{IEEE 754} \rightarrow \epsilon \sim 10^{-16}$$

1.4 ROUND-OFF ERROR (FINITE PRECISION ERROR); CATASTROPHIC LOSS OF PRECISION (C.L.O.P.)

- MULTIPLIES / DIVIDES : NOT MUCH OF A PROBLEM.
FOR "RANDOM" OPERATIONS, EXPECT RELATIVE ERROR TO GROW LIKE $\sqrt{n}$ WHERE n IS THE NUMBER OF OPERATIONS
- ADDITION / SUBTRACTION : SERIOUS PROBLEM WHEN SUBTRACTING $x, -y$ WITH $x \approx y$

$$\begin{array}{r} 3.6326 \\ - 3.6325 \\ \hline 0.0001 \end{array} \quad \begin{array}{l} 5 \text{ SIG. FIGS} \\ 5 \text{ SIG. FIGS.} \\ 1 \text{ SIG. FIG. !} \end{array}$$

1.4.1 EXAMPLE: NUMERICAL DIFFERENTIATION

$$f(x+h) = f(x) + hf'(x) + \frac{1}{2}h^2 f''(x) + \underbrace{O(h^3)}_{\text{ERROR TERM}}$$

$$\frac{f(x+h) - f(x)}{h} = f'(x) + O(h)$$

USE THIS TO NUMERICALLY APPROXIMATE $f'(x)$ AND CONSIDER LIMIT $h \rightarrow 0$

- WITH EXACT ARITHMETIC, ERROR DECREASES MONOTONICALLY AS $h \rightarrow 0$
- WITH FL. PT., ERROR DECREASES ONLY UP TO PT WHERE EFFECTS OF C.L.O.P. BECOME

SIGNIFICANT



2. STABILITY / NUMERICAL STABILITY

- CONSIDER SOME PROCESS / ALGORITHM / MAPPING

$$U(x) \rightarrow y \quad \text{EXACT}$$

$$\hat{U}(\hat{x}) \rightarrow \hat{y} \quad \text{FLOATING PT.}$$

$\hat{x}, \hat{y}$ TYPICALLY VECTORS

HEURISTIC (NON-RIGOROUS) DEFINITION OF STABILITY

CONTINUOUS: IF $\|x - x'\|$ IS SMALL THEN SO IS $\|U(x) - U(x')\|$ $\|\cdot\|$ IS SOME NORM, I.E. SOME DISTANCE BETWEEN x, x' $U(x), U(x')$ (NORMS COULD BE DIFFERENT)

FLOATING PT: IF $\|\hat{x} - \hat{x}'\|$ IS SMALL THEN SO IS $\|\hat{U}(\hat{x}) - \hat{U}(\hat{x}')\|$

- UNFORTUNATE FACT: STABILITY OF $U(x)$ DOES NOT GUARANTEE STABILITY $\hat{U}(\hat{x})$

IF $\hat{U}$ IS NOT STABLE, ROUND OFF ERROR (FP) WILL LIKELY RENDER THE METHOD USELESS

$\hat{U}$ IS THEN CALLED "NUMERICALLY UNSTABLE"
"ILL-CONDITIONED"